

Logarithm Rules Cheat Sheet

Complete logarithm rules cheat sheet with product rule, quotient rule, power rule, change of base, and natural log properties. Free PDF download.

Basic Properties

NAME	FORMULA	NOTES
Definition	$\log_b(x) = y \Leftrightarrow b^y = x$	b is base, x is argument
Log of 1	$\log_b(1) = 0$	Any base raised to 0 is 1
Log of Base	$\log_b(b) = 1$	Base raised to 1 is itself
Log of Power	$\log_b(b^x) = x$	Inverse property
Exponential of Log	$b^{\log_b(x)} = x$	Inverse property

Logarithm Laws

NAME	FORMULA	NOTES
Product Rule	$\log_b(xy) = \log_b(x) + \log_b(y)$	Log of product = sum of logs
Quotient Rule	$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$	Log of quotient = difference of logs
Power Rule	$\log_b(x^n) = n \log_b(x)$	Exponent moves to front
Root Rule	$\log_b(\sqrt[n]{x}) = \frac{\log_b(x)}{n}$	nth root = divide by n

Change of Base

NAME	FORMULA	NOTES
Change of Base Formula	$\log_b(x) = \frac{\log_c(x)}{\log_c(b)}$	Convert to any base c
Using Natural Log	$\log_b(x) = \frac{\ln(x)}{\ln(b)}$	Convert using ln
Using Common Log	$\log_b(x) = \frac{\log(x)}{\log(b)}$	Convert using log ₁₀
Reciprocal Rule	$\log_b(a) = \frac{1}{\log_a(b)}$	Flip base and argument

Natural Logarithm (ln)

NAME	FORMULA	NOTES
Definition	$\ln(x) = \log_e(x)$	$e \approx 2.71828$
ln(1)	$\ln(1) = 0$	
ln(e)	$\ln(e) = 1$	
ln(e ^x)	$\ln(e^x) = x$	
e ^{ln(x)}	$e^{\ln(x)} = x$	

Common Logarithm (log₁₀)

NAME	FORMULA	NOTES
Definition	$\log(x) = \log_{10}(x)$	Base 10 logarithm
log(1)	$\log(1) = 0$	
log(10)	$\log(10) = 1$	
log(10 ⁿ)	$\log(10^n) = n$	
log(100)	$\log(100) = 2$	